ENGR 392 – Reflection Paper

The Impact of AI Hiring Algorithms on Fair Employment Practices

Due: Monday, April 13th, 2026

Team Members

Daniel Lam - 40248073

Juan-Carlos Sreng-Flores - 40101813

Gianluca Girardi - 40228370

Mouhamed Kairson Coundoul - 40248237

Danny Mousa - 40226912

Impact of Technology on Society

Professor Jonathan Wald

1. Concise summary of the problem

The main problem our project looks at is that hiring is becoming less human and more controlled by automated systems. What used to be a process where employers reviewed candidates is now turning into a kind of AI arms race. On one side, companies use hiring algorithms to sort through a large number of applications and on the other side, applicants use generative AI to quickly produce tailored resumes and cover letters. Our topic proposal describes this as algorithmic escalation, where both sides rely more and more on AI. This creates a system where it makes it harder to tell who is actually qualified and instead rewards the candidates that are better at using AI tools.

What makes this a serious issue is that it affects how people get access to jobs, income and future opportunities. If the hiring process becomes unfair, then the effects don't stop at just one rejected application but instead affects a person's career path, confidence and opportunities. The research shows that AI hiring systems make these problems worse because they work like a black box. That means people are judged by systems they do not understand, using rules they cannot see, question or appeal, thus making the whole process unfair.

Another major problem is bias. AI is often presented as neutral and objective but they learn from past hiring data which can reflect unfair patterns from society. If older data was biased toward certain groups then the algorithms can repeat these patterns in new decisions. The research points out that bias can come from missing information, unbalanced data or even how the model is designed. So even if the goal is efficiency, it can lead to more discrimination.

As such, automation does not automatically remove unfairness. Research shows that people see AI-led resume screening as less fair and fairness is not only about whether the system was statistically accurate, it is also about whether people feel they were treated reasonably. If not, trust in the hiring process drops and frustration increases. Overall, the real problem is that hiring values speed and automation rather than transparency and fairness.

2. Concise summary of the proposal

The solution proposed for this problem is to create the FairHire Alliance certification starting in Montreal. FairHire Alliance is an independent certification company that specializes in enforcing certain levels of hiring policies on companies while also providing proper tools and guidance for candidates to demonstrate their capabilities. The organization's goal is to bridge the gap between candidates and companies to provide a more trustworthy environment for hiring fairness, transparency, explainability and accountability. The company's vision is also to redefine the concept of fairness in hiring from a social construct perspective rather than treating it as only as a technological problem.

This solution solves the main issue where companies have currently no incentive to enforce fair hiring practices. Certified companies will benefit from many services the FairHire Alliance (FHA) has to offer. Notably, deeper understanding of their selected candidates and increasing cultural and gender diversity within their teams. They will also be considered more trustworthy on applicants' fair assessment, hence providing more appealing jobs that include the FHA certification logo. The FHA certification logo can be used for various marketing strategies for job posts, events, career fairs, etc. As a tradeoff for these benefits, companies will employ rigorous techniques to provide a more fair and explainable assessment of their candidates. The enforced hiring practices include: balanced structures in decision power, maintaining strict transparency, integrating diversity metrics thoughtfully, increasing stakes of job descriptions for better decision making.

The balanced structure practice involves enforcing a 50/50 or 70/30 human-AI decision power distribution. This hiring practice is mainly to provide confidence that the candidates' applications were decided by a combined decision power without over reliance on AI. Additionally, specific diversity guidelines will be presented to hiring managers. These guidelines may include some company goals in specific metrics and where hiring managers will intuitively spend more time reasoning on the selection of candidates. Some diversity related explanations will be added to their candidature as well and will be displayed in forms of statistics on specific areas where they positively or negatively impact.

It is also important to bring awareness to hirers of how important their decisions impact the company, and as a result, increasing the stake of job descriptions is also often a proper technique to provide more accountability and explainability of their decisions. The candidate selections will also be provided with explanations in comparison to the rejected ones, and proper feedback will be provided to all candidates.

Furthermore, the organization plans to engage with the general population looking for jobs by providing content, courses, and other kinds of help to assist candidates in understanding the selection process that is being enforced. This provides transparency, explainability, and accountability for candidates who are applying to certified companies. It will also help candidates understand where to provide more emphasis in their applications and set up a foundation for better understanding of their skills.

The FairHire organization plans to address over-reliance on LLMs for both companies and candidates. As mentioned earlier, companies will have a balanced structure, however, candidates also need to be demonstrating true potential. As a result, live/in-person interviews and tangible projects or accomplishments walkthroughs will be considered heavily in the selection process although it is already a current standard in many companies.

3. Explanation of how your research led to your proposal

The research guided the proposal to the creation of a FairHire certification by highlighting major issues in Artificial Intelligence (AI) driven hiring systems. In general, the literature emphasizes over and over again that this technology is not inherently fair and requires some sort of oversight. In this context, fairness can be understood as having 2 dimensions:

objective vs subjective fairness. On one hand, the actual unfairness stems from biases reproduced or amplified in the datasets on which the models were trained on. This issue leads to unequal outcomes for certain groups. On the other hand, the perceived unfairness can be explained by the lack of transparency experienced by the candidates during the recruitment process due to their awareness of limited human involvement which leads to unclear final decision making. The findings suggest several factors such as defining clear company policies (transparency), providing diversity related information such as representation statistics within the company (care) and role context which affects the level of attention with which candidates are evaluated (responsibility).

Ling and al. examine how applicants perceive fairness in a balanced hiring process involving human and AI decision making. When conducting their experiment, they learned that applicants responded more positively with screening structures that had higher levels of human involvement, such as 70/30 or 50/50 human-AI decision power distributions[1]. It was examined that predominantly AI structures were negatively perceived. One clear issue that can be taken away from this article is that AI cannot be the sole decision maker, there exists a need for human involvement in the recruitment process.

The authors of *Augmenting diversity in hiring decisions with artificial intelligence tools* demonstrated that diversity related factors such as organizational guidelines and diversity related information greatly influence hiring decisions. Wilkens and al. discovered that when these two factors were ignored, 25% of the experiment's participants selected a female candidate thus creating an inequality. However, when such factors were taken into consideration, the selection of female candidates increased, see Table X. The article demonstrates that without applying these factors, decision makers are prone to simply accept the AI's ranking system without thinking about diversity. This experiment is a concrete example of why a clear definition of company policy accompanied by diversity related factors and job context need to be pieces of the final puzzle.

Table X. Diversity-enhancing hiring decisions (%) pulled from *Augmenting diversity in hiring decisions with artificial intelligence tools.*

Diversity Guidelines	AI Diversity Explanations	Low-Stakes Job (Quantitative)	High-Stakes Job (Qualitative)
Absent	Absent	25%	64%
Absent	Present	48%	80%
Present	Absent	59%	74%
Present	Present	68%	78%

Next, AI has limitations in achieving fairness. In *A Comprehensive Review of AI Techniques for Addressing Algorithmic Bias in Job Hiring,* Albaroudi and al. explains that AI tools reflect unfair patterns. More in depth, these various forms of biases, including representation and measurement bias come from a technical standpoint such as model limitations and the data they were trained on [3]. It is claimed that these problems can be reduced if developers are actively trying to fix them. In other words, there needs to be a will for those in charge. However, Hughes and al. extend this argument by stating that these biases are not purely technical. *Problematizing the role of artificial intelligence in hiring and organizational inequalities: A multidisciplinary review* demonstrates that are rooted in broader social constructs such as race, gender and class. They highlight that different disciplines conceptualize “bias” and “fairness” differently and that the main actors involved in AI are computer science related. The authors critique the idea of using AI as a technofix and that it can in turn amplify these social inequalities[4]. Together, these findings suggest that achieving fairness requires more than technical improvements. It reinforces the claim that solutions must be supported by organizational oversight and clearly defined values.

In the end, the literature points to a clear conclusion: the two dimensions of fairness cannot be left to algorithms alone but must be supported by values such as transparency, care and responsibility. Across the articles, it has been repeatedly shown that biases are reproduced by training data and reshaped by social constructs, while fairness is determined by the level of human involvement and transparency in the decision making process. Additionally, factors such as contextual information and diversity guidelines have an impact in shaping hiring outcomes. Together these findings reinforce the need for structural oversight, laying out the foundation for the FairHire certification approach. In other industries, such as food certifications for gluten free, halal and fairtrade have had the ability to build consumer trust by improving transparency and guaranteeing compliance. Research in this industry has shown that there is a need to "*improve transparency and authenticity*"[5] in complex systems, suggesting that similar mechanisms could be applied to hiring systems. The certifications provide assurance to the consumers while enforcing company policies. In other words, there is a need to have better "*labeling and traceability*"[5] indicating that the adoption of certifications plays a key role in shaping consumer trust and awareness. White and al. mention that such labels have shown to "*simplify complex information*"[5] which is extremely relevant as AI hiring systems are difficult to interpret. This is where FairHire steps in, as it has the mission to apply the proposed solution to fix the bridge of trust between applicants and hiring companies

However there exists a flaw, there are no incentives for companies to comply with the policies and obtain a FairHire stamp of approval. In order to be effective, proper marketing and client acquisition measures need to be put into place. Without incentives for adoption, the idea of a certification will not achieve meaningful impact.

4. Explanation of why group chose a particular presentation format

As part of planning our presentation, it became necessary to make a choice whether to create a printed or digital presentation. We chose the latter option as it allows more freedom to make changes and adjustments until the very last moment before the actual delivery. Using a poster would force us into making the final version way too early since it would already be physically available and unable to be changed once completed.

Secondly, presenting our ideas in the form of a poster was convenient as far as our topic is concerned. Indeed, considering FairHire Alliance to be a real certification program, it allowed us to give it a more tangible form. Thus, we designed a logo, chose a certain color palette, and presented our ideas in such a way that the resulting design looked like what a real company would present. In other words, this method gave our project a more authentic look similar to what one could expect from a marketing campaign at a career fair.

Thirdly, using only a five-minute period for presentation, the format of a poster proved to be convenient since everything would fit on one page. In other words, instead of showing several slides in a row, we would be able to go through the whole material without having to pause between them in order to click on the next slide.

5. Brief summary of contributions of each group member

References:

[1] U. Wilkens, I. Lutzeyer, C. Zheng, A. Beser, and M. Prilla, “Augmenting diversity in hiring decisions with artificial intelligence tools”, *The International Journal of Human Resource Management*, vol. 36, no. 9, pp. 2585–2622, Apr. 2025, doi:10.1080/09585192.2025.2492867.

[2] B. Ling, B. Dong, and F. Cai, “Applicants’ Fairness Perception of Human and AI Collaboration in Resume Screening,” International Journal of Human-Computer Interaction, pp. 1–12, Dec. 2024, doi: https://doi.org/10.1080/10447318.2024.2437235.

[3] K. D. Hughes, A. Konnikov, N. Denier, and Y. Hu, “Problematizing the role of artificial intelligence in hiring and organizational inequalities: A multidisciplinary review,” Human Relations, Dec. 2025, doi: https://doi.org/10.1177/00187267251403902.

[4] E. Albaroudi, T. Mansouri, and A. Alameer, “A comprehensive review of AI techniques for addressing algorithmic bias in job hiring,” *AI*, vol. 5, no. 1, pp. 383–404, 2024, doi:10.3390/ai5010019.

[5] G. R. T. White and A. Samuel, “Fairtrade and Halal Food Certification and Labeling: Commercial Lessons and Religious Limitations,” *Journal of Macromarketing*, vol. 36, no. 4, pp. 388–399, 2016, doi: 10.1177/0276146715620236.